

C64 Keyboard: Which Cables Do I Need?

You can find three cable drawings in the /Cables/pdf subfolder:

- **KB_Cab.pdf:** The 20-way ribbon cable connecting the keyboard to the keyboard connector PCB.
- **KSw_Cab.pdf:** The cable connecting the Kernal EPROM adapter PCB to /RESET and /EXROM.
- **Ulti_Cab.pdf:** The cable connecting the keyboard to the "Buttons" pin header on the Ultimate 64 Elite II or C64 Ultimate mainboard.

Which cables you need depends on how you intend to use the keyboard.

- **USB keyboard for VICE:** No cables required.
- **C64 keyboard without Kernal switching or reset functionality:** KB_Cab.
- **C64 keyboard with Kernal switching and reset functionality:** KB_Cab + KSw_Cab.
- **Keyboard connected to an Ultimate 64 Elite II or C64 Ultimate:** KB_Cab + Ulti_Cab.

So, you never need more than two cables.

Making the ribbon cable

Making a ribbon cable requires no soldering and should only take a few minutes. All you need is a vise to press the connector onto the cable and a pair of scissors to cut the ribbon cable to length.

Making the Dupont cables

For the Dupont cables, you can either use a ready-made 40 cm Dupont ribbon cable with individual female connectors, which can be used as is, or additionally use a Dupont connector assortment with crimp housings in the required sizes (3-way, 4-way and 5-way). The latter option is a bit better, as the housings keep the wires in the correct order and prevent mix-ups. To do this, you can remove the crimp terminals from the individual 1-way housings (pointy tweezers) and insert them into the appropriate multi-way housings from the assortment.

If you want to crimp the connectors yourself, I recommend the **IWISS SN-025 crimping tool**. It's the only tool I've personally found that can properly crimp Dupont terminals. Other tools, such as the SN-28B or SN-48B, are sometimes advertised as Dupont crimping tools, but they aren't suitable for this purpose. They're good tools, just not for Dupont terminals.